

AI, Video Limits & Database Storage Analysis

1. Language Model / AI Usage - Cost Analysis

Current AI Implementation: The platform uses basic AI features (no external LLM API calls). See proof:

File Path: server/services/adminConfigService.js:4-7

```
const DEFAULT_CONFIG = {
  feature_flags: {
    ai_summaries: true,
    auto_credit: true,
    bulk_approvals: false,
    system_broadcasts: true,
  },
```

NO OpenAI/GPT API Integration Found in Codebase!

I searched for:

- openai - NOT FOUND
- gpt- - NOT FOUND
- api.openai.com - NOT FOUND
- azure openai - NOT FOUND

Conclusion: Your current AI features are RULE-BASED logic, not external LLM API calls. **Cost = \$0** for current AI.

2. Video Duration Limit - 60 Seconds

File Path: server/services/adminConfigService.js:11-26

```
plan_limits: {
  free: {
    partner_limit: 5,
    search_daily: 20,
    request_limit: 3,
    product_limit: 20,
    video_limit: 2,      // FREE: 2 videos max
    agent_limit: 10,
  },
  premium: {
    partner_limit: 50,
    search_daily: 200,
    request_limit: 50,
    product_limit: 500,
    video_limit: 200,    // PREMIUM: 200 videos max
    agent_limit: 999,
  },
},
```

Video limit is COUNT-based, not duration-based!

- FREE: 2 videos
- PREMIUM: 200 videos

No 60-second duration enforcement found in code. If you need to add duration limit:

```
// Should add in upload service:
// MAX_VIDEO_DURATION_SECONDS = 60

function validateVideoUpload(file) {
  const MAX_DURATION = 60 // seconds
  // Check video metadata.duration...
  if (videoDuration > MAX_DURATION) {
    throw new Error('Video must be 60 seconds or less')
  }
}
```

3. Image Uploads - Database Storage Estimate

File Path: server/services/adminConfigService.js:11-26

```
free: {
  product_limit: 20,    // 20 products
  video_limit: 2,
},
premium: {
  product_limit: 500,   // 500 products
  video_limit: 200,
  agent_limit: 999,
},
```

Image Storage Estimates:

Plan	Products	Max Images/Product	Total Images	Estimated Storage
Free	20	10	200	~500 MB
Premium	500	10	5,000	~12.5 GB

Per Image Calculation:

- Average product image: 100-500 KB (compressed JPEG)
- With thumbnails: +50 KB/image
- Total per image: ~150-550 KB

4. Database Storage Requirements

Data Files (JSON-based):

```
// Storage files used:
- users.json
- user_profiles.json
- products.json
- buyer_requests.json
- contracts.json
- messages.json
- violations.json
- coupon_codes.json
- coupon_redemptions.json
- traffic_analytics.json
- notifications.json
```

Estimate for 10,000 Users:

Data Type	Est. Records	Est. Size

Users	10,000	~10 MB
Data Type	Est. Records	Est. Size
Products	50,000	~50 MB
Buyer Requests	30,000	~30 MB
Contracts	5,000	~5 MB
Messages	100,000	~20 MB
Media URLs*	50,000	~5 MB

*Note: Actual images/videos stored in cloud (S3/Cloudflare R2), not in database.

5. Cost If Adding External LLM API

If you add OpenAI/GPT later:

Feature	Request	Est. Cost/1K users
Chat summary	50 words	\$0.002
AI negotiation	100 words	\$0.004
Product matching	50 words	\$0.002

Monthly Estimate (1,000 active users, 10 messages each):

- 10,000 summaries × \$0.002 = \$20/month
- Add buffer = ~**\$50-100/month max**

6. Conclusion

Question	Answer
Is AI expensive?	NO - Currently rule-based, \$0 cost
Video 60s limit?	NOT enforced - Currently count-based (2 free / 200 premium)
Image storage?	~500KB/image - Cloud storage needed
Database for 10K users?	~150 MB for JSON files
Adding LLM later?	~\$50-100/month if needed

Recommendations

1. Add duration check: `MAX_VIDEO_DURATION = 60` in upload service
2. Use Cloudflare R2/S3 for media storage (not database)
3. If adding AI later, use GPT-4o-mini for cost savings